The Hexagonality Law in Real Materials IV

Oxides: hexagonal perovskites and battery cathodes

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. Chemists judge oxide structures with two rules of thumb: Pauling's third rule (octahedra that share faces are unstable) and Goldschmidt's tolerance factor (a large A cation pushes a perovskite towards hexagonal stacking). DICTO proposed that three-dimensional stacking rules can replace such rules of thumb. We show that both rules are measured by one number, the hexagonality h of the oxygen stacking (#66). In a hexagonal perovskite ABO_3 of any stacking, each B octahedron shares faces with 2h neighbours and corners with $6(1 - h)$. In a layered battery cathode $LiMO_2$, the lithium–metal octahedra that share a face number 2h per lithium site on average. Measured data follow the law: the volumes of four $BaRuO_3$ polytypes lie on one straight line in h (within 0.2% for the three precisely measured ones), and pressure walks the structure down h, 9R → 4H → 6H → 3C. In Li_xCoO_2 on charge, the H1-3 phase puts all its face-sharing into the empty gaps and keeps lithium only where it shares no face, giving the bound $x \le 1 - h$ for the stable phases; its layer spacing, 4.52 Å, lies on the law line between O3 and O1 (4.525 Å). The law has limits: in $BaMnO_3$ the volume follows it but the energy does not.

1. Why oxides

Papers #67–#71 tested the Hexagonality Law on elements, simple compounds and alloys: there the atoms themselves are the close-packed balls. In most oxides the oxygen ions are the close-packed layers and the metal ions sit in the holes between them, mostly in octahedra. Whether two neighbouring octahedra share a corner, an edge or a face then depends only on the stacking of the oxygen layers, and Pauling's third rule says the face-sharing ones cost energy, because their metal ions come close. DICTO proposed that three-dimensional stacking rules (the Hexagonality Law, the Rule of Halves, the Swap Rule) can stand in for rules of thumb used in chemical modelling. Here we test the first of them on two families where the stacking is the main story: the hexagonal perovskites, and the layered oxides that hold the lithium in rechargeable batteries.

2. The count

Let h be the share of hexagonal layers in the stacking of the close-packed oxygen (or AO_3) layers: a layer is hexagonal when the layers above and below it sit in the same position, A, B or C. An octahedral hole between two layers sits on the third position, the one neither layer uses.

Theorem. (i) In a perovskite ABO_3 built of close-packed AO_3 layers with B in the octahedra formed by oxygen alone, each B octahedron shares a face with 2h neighbours and a corner

with 6(1 − h) neighbours, for any stacking. (ii) In a layered oxide $LiMO_2$ (oxygen layers in pairs around the M layers, lithium in the gaps), the face-sharing lithium–metal pairs number 2h per lithium site; a gap bounded by two cubic layers (O3 type) has none, a gap bounded by two hexagonal layers (O1 type) has two, a mixed gap (O2 type) has one.

Proof. Take one oxygen layer and the octahedra just above and just below it. Each sits on the position missing from its own two layers. If the layer is hexagonal, the layers above and below it are on the same position, so the two octahedra are on the same position too and share the three oxygen atoms of the layer between them: a face. If the layer is cubic, the two octahedra are on different positions and meet in an edge (layered oxides) or, in the perovskite, where only one hole in four is filled, in single corners with three neighbours. Each B octahedron of a perovskite has one layer above and one below; each lithium gap has one bounding layer on each side. Averaging over the stacking gives 2h and 6(1 − h). ∎ The counts were checked from explicit coordinates for 3C, 2H, 4H, 6H and 9R perovskites and for the O3, O1, O2 and H1-3 layered stackings (study files perovskite_counts.py, cathodes.py).

3. Hexagonal perovskites

$BaRuO_3$ is made in four polytypes; the stacking can be read from the interfaces between neighbouring octahedra (C corner, F face) listed by Jin et al.: 9R (CFF repeated, h = 2/3), 4H (CF, h = 1/2), 6H (FCC, h = 1/3) and 3C (all C, h = 0). The face-sharing share equals h in every case, as Theorem (i) requires. Their volumes per formula unit lie on one straight line:

polytype	h	volume measured (Å^3)	law line 64.25 + 6.82h	off
3C	0	64.28	64.25	+0.06%
6H	1/3	65.88 (c known to two figures)	66.52	−0.97%
4H	1/2	67.51	67.66	−0.22%
9R	2/3	68.91	68.79	+0.16%

Since the volume rises with h, pressure favours lower h at first order (#70, Proposition). Jin et al. made the polytypes at increasing pressure: 9R at ambient, 4H at 3 GPa, 6H at 5 GPa, 3C at 18 GPa. The sequence walks down h step by step: 2/3, 1/2, 1/3, 0. The same direction is seen with chemistry: in $Ba_{1-x}Sr_xRuO_3$, the smaller strontium takes the structure 9R → 4H → 3C. The tolerance factor says which way the structure leans; the law says by how much the volume and the face-sharing change at each step.

$BaMnO_3$, computed by Chen and Millis (DFT, antiferromagnetic, per Mn), gives a second case. The volumes of 3C (61.9), 4H (66.1) and 2H (71.0 Å^3) are close to the law: 4H lies 0.5% below the straight line through 3C and 2H. The energies are not: 2H is lowest, 4H is 109 meV higher and 3C 775 meV higher, so 4H lies 278 meV below the line. The barium ion is too large for cubic $BaMnO_3$ (tolerance factor above 1), which makes 3C unusually costly; in the language of #68, the second-neighbour coupling is large here.

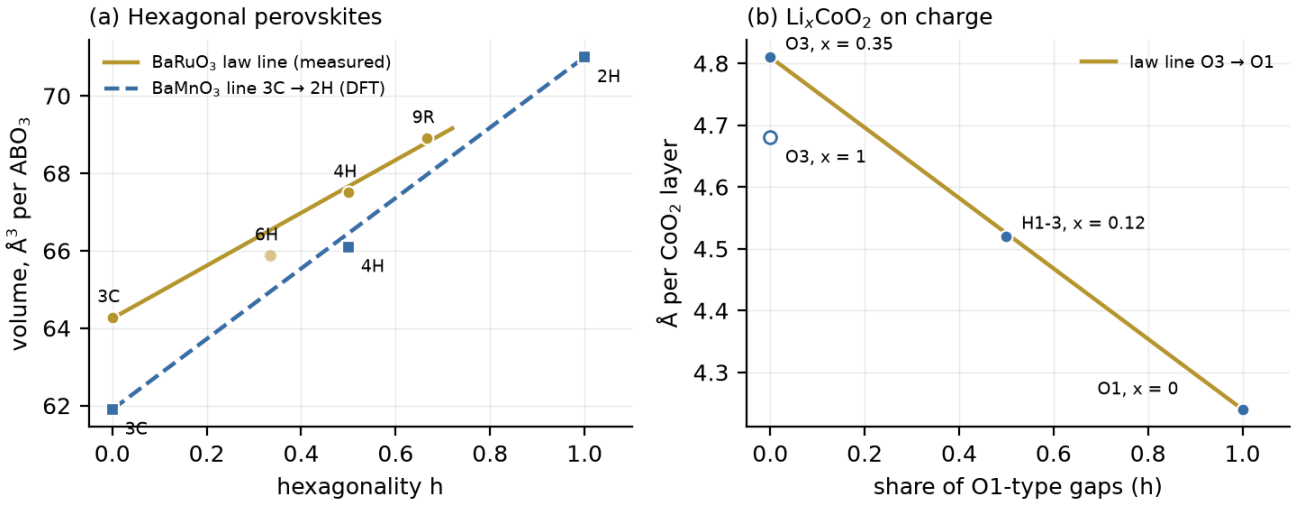


Figure 1. (a) Volume per ABO_3 against the hexagonality h: $BaRuO_3$ measured (Jin et al.; 6H pale, its c known to two figures) with the law line, and $BaMnO_3$ computed (Chen and Millis) with the straight line from 3C to 2H. (b) Layer spacing of Li_xCoO_2 (measured, Motohashi et al.) against the share of O1-type gaps: the H1-3 phase sits on the line from O3 to O1; open circle: O3 with full lithium.

4. Lithium cobalt oxide on charge

$LiCoO_2$, the cathode of most phone batteries, has cubic O3 stacking (h = 0): every lithium octahedron shares only edges with the cobalt octahedra. On charge the lithium leaves. Motohashi et al. measured the full series: O3 down to x = 0.35, then the H1-3 phase at x = 0.12, then CoO_2 with O1 stacking (h = 1) at x = 0. H1-3 has alternating O3-type and O1-type gaps (h = 1/2), and the lithium sits only in the O3-type gaps (Van der Ven et al. predicted this staging; Motohashi et al. confirm the structure). By Theorem (ii) the face-sharing count per gap runs 0, 2, 0, 2, …: H1-3 puts all its face-sharing into the gaps it leaves empty.

> **Proposition (lithium bound).** If lithium occupies only gaps in which it shares no face with a metal octahedron, then $x \le 1 - h_{gap}$, where h_{gap} is the share of O1-type gaps. All stable phases in Motohashi et al.'s series obey it: O3 ($h_{gap} = 0$, $x \le 1$), H1-3 (1/2, x = 0.12), O1 (1, x = 0).

This is the oxide counterpart of the hydride bound of #71 ($h \le 2 - x$): in both, the number of guest atoms is limited by the faces the stacking creates. One known structure breaks it: O2-$LiCoO_2$, in which every gap is mixed and every lithium shares one face with cobalt. It is made only by ion exchange from a sodium cobalt oxide (Delmas, Braconnier and Hagenmuller) and is metastable (Carlier et al.). A face-sharing pair therefore costs energy rather than being forbidden, as Pauling's rule says; the law counts the pairs exactly.

The layer spacing tests the law as a number. Motohashi et al. give 4.81 Å per CoO_2 layer for O3 at x = 0.35–0.50 and 4.24 Å for O1. The law line at h = 1/2 gives 4.525 Å; H1-3 measures 4.52 Å (c/6 = 4.522 Å), within 0.11%. Spacings in staged intercalation compounds are known to add layer by layer (as in graphite intercalation), so this agreement is expected once H1-3 is read as half O3 and half O1; what the law adds is the face count that tells which half holds the lithium.

5. Where the law stops

(1) The $BaMnO_3$ energies are far from the straight line (Section 3), although the volumes are close. (2) The sizes of face-sharing clusters (dimers in 4H, trimers in 9R, endless chains in 2H) are not linear in h; they depend on runs of hexagonal layers and drive the magnetism Jin et al. report. (3) The layer spacing of Li_xCoO_2 also depends on x at fixed stacking (4.68 Å at x = 1 against 4.81 Å at x = 0.35), so the law describes the stacking part only. (4) In nickel-rich cathodes the H2 → H3 change keeps O3 stacking (h = 0 throughout); its large shrinkage of the layer spacing is outside the law.

6. Conclusion

Pauling's third rule and the tolerance factor become one count: the faces shared by oxygen octahedra are 2h per octahedron in perovskites and 2h per lithium site in layered cathodes, for any stacking. Measured volumes of $BaRuO_3$ lie on the law line, pressure walks them down h, and charged $LiCoO_2$ keeps its lithium where the count gives zero, $x \le 1 - h$.

7. Prior art and credit

The stacking notation for hexagonal perovskites (h and c layers, nH and nR) goes back to Katz, Ward and co-workers (Donohue, Katz and Ward); the O/P notation for layered oxides is Delmas, Fouassier and Hagenmuller's; H1-3 as an O1/O3 hybrid with lithium in every other gap is Van der Ven et al.'s prediction and Motohashi et al.'s measurement; that face-sharing costs energy is Pauling's third rule. The counts 2h and 6(1 − h) for any stacking, the split of the face count between gaps, the lithium bound and the comparison with the law line are stated here with proofs; the proofs are short. DICTO proposed replacing chemical rules of thumb with three-dimensional stacking rules; the choice of perovskites and cathodes and the computations are joint work.

> **Sources checked** (28 October 2026). Read: C.-Q. Jin et al., High-pressure synthesis of the cubic perovskite $BaRuO_3$ and evolution of ferromagnetism in $ARuO_3$ (A = Ca, Sr, Ba) ruthenates, PNAS 105, 7115 (2008), PMC2438213 (Table 1, synthesis); L. T. Nguyen and R. J. Cava, Hexagonal perovskites as quantum materials, arXiv 2006.05864 (pp. 1–3, notation and $Ba_{1-x}Sr_xRuO_3$; reference list); arXiv 2107.07165 (4H, 9R cells as quoted); T. Motohashi et al., Electronic phase diagram of the layered cobalt oxide system Li_xCoO_2 ($0.0 \le x \le 1.0$), Phys. Rev. B 80, 165114 (2009), arXiv 0909.3556 (Table I and Section III.1); H. Chen and A. J. Millis, Phase diagram of $Sr_{1-x}Ba_xMnO_3$ as a function of chemical doping, epitaxial strain and external pressure, arXiv 1509.06650 (Table II); AFLOW prototype ABC2_hP8_186_b_b_ab-001 (O2-$LiCoO_2$). Abstract only: A. Van der Ven, M. K. Aydinol, G. Ceder, G. Kresse and J. Hafner, Phys. Rev. B 58, 2975 (1998) (repository copy); arXiv 2212.14667. Cited as given in those sources, not read: P. C. Donohue, L. Katz and R. Ward, Inorg. Chem. 5, 335 (1966); the 6H $BaRuO_3$ cell (c ≈ 14 Å) as quoted second-hand; C. Delmas, C. Fouassier and P. Hagenmuller, Physica B 99, 81 (1980); C. Delmas, J.-J. Braconnier and P. Hagenmuller, Mater. Res. Bull. 17, 117 (1982); D. Carlier et al., On the metastable O2-type $LiCoO_2$, Solid State Ionics 144, 263 (2001); G. G. Amatucci, J.-M. Tarascon and L. C. Klein, J. Electrochem. Soc. 143, 1114 (1996). Web searches: LixCoO2 H1-3 O1 lattice parameters; O2-LiCoO2 Carlier Delmas; BaIrO3 high-pressure polytypes (Cheng et al. 2009, cell volumes not reachable); Katz and Ward hexagonal perovskite stacking (the 1964 paper not reached); Van der Ven 1998 abstract.

> **Data and code.** Study 2026-10-28-perovskites-cathodes (baruo3.py, bamno3.py, cathodes.py, perovskite_counts.py, fig72.py, with outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275; The Hexagonality Law in Real Materials II (#70), 10.5281/zenodo.23723565; The Hexagonality Law in Real Materials III (#71), 10.5281/zenodo.23726064.